MIL-M-63321A(AR)
8 April 1981
SUPERSEDING
MIL-M-63321(AR)
24 August 1979

MILITARY SPECIFICATION

MICROCIRCUIT, DIGITAL, CMOS (FINAL LOGIC)

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains requirements not covered by the drawings and provides quality assurance provisions for fabrication and packing of a complementary, metal oxide, semiconductor (CMOS) integrated circuit used in the Mine, Antipersonnel: HE, M74 and Mine, Antitank: HE, M75.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510 - Microcircuits, General Specification for
MIL-M-55565 - Microcircuits, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

9292963 - Microcircuit, Digital, CMOS (Final Logic)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Device classification. The microcircuit (device) shall conform to the product assurance level class B of MIL-M-38510 and as specified herein.

3.2 Device. The device shall comply with all requirements specified on Drawing 9292963 and with all requirements specified in applicable specifications and standards.

3.3 Design and construction. The design and construction shall comply with the requirements of MIL-M-38510 for class B devices and 3.2 herein.

3.3.1 Glassivation. The microcircuit shall be glassivated with silicon dioxide to render it impervious to a contaminated environment. The glassivation shall cover all electrical conductors except the bonding pads.

3.4 Environmental and mechanical. Devices to be delivered under this specification shall be capable of meeting the following MIL-STD-883 environmental and mechanical tests when the devices are subjected to these tests in accordance with this specification.

Environmental Tests	Method	Conditions
Moisture resistance	1004	
Steady state life	1005	D, 1000 hrs @ 125°C
Temperature cycling	1010	C, 10 cycles
Seal (Fine & Gross)	1014	(A or B & C respectively)
High temperature storage	1008	C
Burn-in	1015	D, 168 Hours @ 125°C

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Mechancial Tests	Method	Conditions
Mechanical shock	2002	D
Solderability	2003	
Lead integrity	2004	B2
Resistance to solvents	2015	
Vibration, variable frequency	2007	A
External visual	2009	
Internal visual	2010	B
Bond strength	2011	D
Physical dimensions	2016	
Constant acceleration	2001	E, Y_1 orientation only

3.5 Electrical characteristics. The device shall have the following electrical characteristics.

3.5.1 Operating ratings. See Table I.

TABLE I. -- OPERATING RATINGS

CHARACTERISTICS	SYMBOL	RATING	UNIT
DC Supply Voltage	V_{DD}	-0.5 to 7.5	Volts
All Inputs	V_{IN}	V_{IN} _ V_{DD}	Volts
Temperature			
Operating 1/	$T_{A(OP)}$	-45 to +85	°C
Storage	$T_{A(STG)}$	-55 to +125	°C

1/ Devices shall be capable of Burn-in at +125°C (See 4.5.16)

3.5.2 Performance characteristics. The device shall meet the performance characteristics of TABLE II.

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TABLE II. ELECTRICAL PARAMETERS.

CHARACTERISTICS	PINS	V_{DD}	T_A 1/	MIN	MAX	UNIT	2/	3/
Input Current	2, 3, 4, 5, 6, 7, 9, 10, 11, 13	7.5			100	nA	3.5.2.1	4.5.4
	2	5.0		0.50	5	uA	3.5.2.2	4.5.7
	11	5.0		0.50	5	uA	3.5.2.3	4.5.8
	9, 10	5.0		0.75	5	uA	3.5.2.4	4.5.8
Leakage Current (Oscillator)		7.5			35	uA	3.5.2.5	4.5.5
Leakage Current Low		7.5	+25°C		4	uA	3.5.2.6	4.5.6
		7.5			10	uA	3.5.2.7	4.5.6
Output Low @ (1.0)v	1, 15	5.0		1.25		mA	3.5.2.8	4.5.9
	12, 14	5.0		100		uA	3.5.2.9	4.5.9
Output High @ (V_{DD} - 1.0)v	1, 15	5.0		50		uA	3.5.2.10	4.5.10
	12, 14	5.0		100		uA	3.5.2.11	4.5.10
Switching Point	13	7.5		2.5	5.0	V	3.5.2.12	4.5.11
Noise Immunity	2, 3, 4, 5, 6, 7, 9, 10, 11	5.0		1.0		V	3.5.2.13	4.5.12
		7.5		1.5		V	3.5.2.14	4.5.12
Functional Characteristics							See 3.5.3	4.5.13

1/ Unless otherwise specified, the MIN/MAX values apply over the temperate range $-45°C \leq T_A \leq +85°C$.
2/ Requirement paragraph (assigned in this table).
3/ Test method paragraph.

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3.5.3 Functional operation. Unless otherwise specified, the input signals used during the functional tests shall have the following characteristics:

Rise time	(t_r)	1.0 microsecond (max)
Fall time	(t_f)	1.0 microsecond (max)
Time	(high)	10 microsecond (min)
Voltage	(low)	1.0 volts max @ V_{DD} = 5.0 volts
		1.5 volts max @ V_{DD} = 7.5 volts
Voltage	(high)	4.0 volts min @ V_{DD} = 5.0 volts
		6.0 volts min @ V_{DD} = 7.5 volts

3.5.3.1 Self-destruct logic. The device shall conform functionally to the timing diagram shown in Figure 1 and 2, when tested as specified in 4.5.13.

3.5.3.2 Cycle monitor logic. The device shall conform functionally to the timing diagram shown in Figure 3, when tested as specified in 4.5.14.

3.5.3.3 Target sensor logic. The device shall conform functionally to the timing diagram shown in Figure 4, when tested as specified in 4.5.15.

3.5.3.4 Oscillator check logic. The device shall conform functionally to the timing diagram shown in Figure 5, when tested as specified in 4.5.13.

3.5.4 Oscillator function (Pins 12, 13, 14). When the device is connected as shown in Figure 10 it shall function as an oscillator such that the output signal at pin 14 (F_{23}) meets the requirements specified in a, b, and c below. These requirements shall be tested for as specified in 4.5.14.

a. When the temperature varies over the operating range (-45°C to +85°C) and the voltage V_{DD} varies from 3.5 to 4.5 volts, the period of the output signal at pin 14 shall not shift more than ± 5% from the output signal period measured at +25°C and with V_{DD} equal to 4.0 volts.

b. When the temperature varies over the operating range (-45°C to +85°C) and the voltage V_{DD} varies from 5 to 7.5 volts, the period of the output signal at pin 14 shall not shift more than ± 5% from the output signal period measured at +25°C and with V_{DD} equal to 6.2 volts.

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c. When the temperature is +25°C and V_{DD} is 7.0 volts, the period of the output signal at pin 14 shall be 11 ± 2.5 milliseconds.

3.5.5 Dual pins/bond pads. The Q and $\overline{C}$ lines on the microcircuit shall each be accessible to external device signals via dual bond pads. Each bond shall be connected to its own package pin. For the Q line these pins are 5 and 6, and for the $\overline{C}$ line these pins are 3 and 4. When tested as specified in 4.5.15, the maximum resistance between pins 5 and 6, and between pins 3 and 4 shall be 10^4 ohms.

3.5.6 Screening. All devices to be delivered in accordance with this specification shall have been subjected to the screening procedures as specified in MIL-STD-883, Method 5004 for class B devices, and as specified herein.

3.5.6.1 Burn-in. The device shall be subject to the burn-in test as specified in MIL-STD-883, Method 5004 and 4.5.16 herein.

3.5.7 Steady state life. The device shall be subject to the steady state life test of MIL-STD-883, Method 1005, Condition D and 4.5.17 herein.

3.6 Marking. The device shall be marked with the information specified in paragraphs 3.6.1 thru 3.6.5, and the primary markings (index point, part number, and inspection lot identification code) shall be located on the top surface of the package. The other markings may be placed in any suitable location so as to perform their functions and not interfere with the primary markings.

3.6.1 Index point. See 3.2.

3.6.2 Part number. The complete part number shall be as follows:

9292963X

where 9292963 is the Drawing number and X the drawing revision letter, if applicable.

3.6.3 Inspection lot identification code. The inspection lot identification code marking shall be in accordance with MIL-M-38510.

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3.6.4 Manufacturer's designating symbol. The manufacturer's designating symbol marking shall be in accordance with MIL-M-38510.

3.6.5 Other markings. The vendor may add additional markings for his own use on the bottom surface of the package provided that these markings do not interfere with the required markings.

3.7 Data requirements. Data shall be recorded and retained for all tests called out by this specification in accordance with 4.5.19.

3.8 Process controls. All process controls essential for the manufacturing of microcircuits to assure a reliable end item shall be submitted to the designated agency specified in 6.5 herein, prior to the start of fabrication.

3.8.1 Metallization. At random intervals, the Government inspector shall monitor the contractor's quality and acceptability of metallization on the semiconductor devices.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.10 Workmanship. See MIL-M-38510.

3.11 Country of origin. The device shall be manufactured and sold by a qualified producer in the United States or in another country which is a party to one of the existing international agreements ratified by the United States, such as the NATO STANAG-4093, Mutual Acceptance of Qualification Approvals for Electronic Parts and the United States/Canada/Australia Agreement for Qualification of Products for Non-resident Manufacturers.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-M-38510 shall apply and are hereby made part of this detail specification. The supplier is responsible for the performance of all inspection requirements as specified herein, and the supplier may use his own or any other facilities suitable for the performance of the tests and inspection requirements specified herein, unless disapproved by the Government. In addition, the Government reserves the right to witness or to perform any of these tests and inspections set forth in this specification where such tests and inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The Government reserves the right to audit the data resulting from the manufacturer's performance of these tests and inspections.

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4.2 Classification of inspection. The following types of inspection shall be conducted on this item.

a. First Article Inspection.

b. Quality Conformance Inspection.

c. Screening

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article inspection lot size after screening shall consist of fifty (50) devices, minimum.

4.3.2 Inspections to be performed. First article inspection shall be in accordance with the applicable requirements of MIL-M-38510 and as modified by this specification, except that the procuring agency shall be the qualifying activity. First article requirements shall consist of screening as per 4.4.2.1 and inspections specified in subgroups "a" through "g" listed in TABLE III. First article items may be subjected by the qualifying activity (Government) to all of the examinations and tests specified in this specification and to any requirements of the applicable drawings.

a. Screening. Screening shall be performed on all devices in accordance with the inspection flow specified in 4.4.2.1.

4.3.3 Rejection. If any device fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a device in the sample to comply with any of the stated requirements.

4.3.4 Requalification. The device shall be subjected to another First Article as stipulated in the contract if:

a. The contractor's manufacturing techniques or process controls which affect the form, fit, function or internal geometry of the device are revised by either the contractor or one of his sub-contractors supplying parts or assemblies.

b. The contractor relocated his assembly facilities.

c. The contractor uses a new sub-contractor for material, parts, or assembly.

d. A change occurs in devices's material, performance or assembly.

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TABLE III. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63321A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
	MICROCIRCUIT, DIGITAL, CMOS (FINAL LOGIC)	1 OF 2	9292963
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Screening	Note 1		3.5.6	4.4.2.1
	Subgroup a	25			
	Electrical parameters @ +85° ± 5°C, -45°C ± 5°C			3.5.2	Table II, 4.4.3.1
	Oscillator function			3.5.4	4.5.14
	Dual pin/bond pads			3.5.5	4.5.15
	Subgroup b	25			
	Mechanical shock			3.4	MIL-STD-883, Method 2002, D
	Vibration, variable frequency			3.4	MIL-STD-883, Method 2007, A
	Moisture resistance			3.4	MIL-STD-883, Method 1004
	Seal (fine leak)			3.4	MIL-STD-883, Method 1014, A or B
	Seal (gross leak)			3.4	MIL-STD-883, Method 1014, C
	External visual			3.3,3.4, 3.6,3.10	MIL-STD-883, Method 2009
	Electrical parameters @ 25° ± 5°C,			3.5.2	Table II, 4.4.3.1
	Country of origin			3.11	Visual
	Subgroup c	25			
	Resistance to solvents	3		3.4	MIL-STD-883, Method 2015
	Bond strength (bond pulls Note 2)			3.4	MIL-STD-883, Method 2011, D
	Subgroup d	25			
	Solderability (leads inspected Note 3)			3.4	MIL-STD-883, Method 2003

NOTES: 1. See Sheet 2.

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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TABLE III. FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63321A(AR)

PARAGRAPH	TITLE		SHEET		DRAWING NUMBER
	MICROCIRCUIT, DIGITAL, CMOS (FINAL LOGIC)		2 OF 2		9292963
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Subgroup e	8			
	Static charge protection circuit test			3.2	4.5.18
	Subgroup f	2			
	Lead integrity			3.4	MIL-STD-883, Method 2004,B2
	Seal (fine leak)			3.4	MIL-STD-883, Method 1014, AorB
	Seal (gross leak)			3.4	MIL-STD-883, Method 1014, C
	Subgroup g	25			
	Steady state life			3.5.7	4.5.17
	Electrical parameters @ 25°C ± 5°C			3.5.2	Table II, 4.4.3.1

NOTES:
1. Devices which fail any test criteria in the screening sequence shall be removed from the First Article Inspection Lot. The contractor should choose a First Article Inspection lot size that is adequate for the needs of the First Article Inspection.
2. The sample size is the number of bond pulls selected from a minimum number of 10 devices.
3. The sample size is the number of leads inspected. In no case shall less than 3 devices be used to provide the number of required leads.
4. All electrical parameters shall be recorded as go/no-go data.
5. The same sample units shall be subjected to all the tests in a subgroup.
6. Devices used in subgroup a shall be used in subgroup b. Devices used in subgroups a and b may be used in subgroups c, d, e and f.
7. Test within each subgroup shall be performed in the sequence shown.
8. First Article Sample: 50 Units -- 25 Units for groups a - f, 25 Units for group g.

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. Inspection lot formation shall be as designed in MIL-M-38510 for Class B devices.

4.4.2 Examinations

a. Quality conformance inspection. Quality conformance inspection shall consist of the inspections specified for Group A and Group B and when required Group C. Quality conformance inspection requirements shall be in accordance with the Classification of Defects and Test paragraph 4.4.2.1 through 4.4.2.4.

b. Inspection requirements. Inspection requirements for this device shall be in accordance with the requirements of MIL-M-38510 and MIL-STD-883, Method 5005 for Groups A, B, and C tests except as modified herein. See 4.4.2.1 through 4.4.2.4.

c. Screening shall be performed on all devices in accordance with the inspection flow specified in 4.4.2.1.

d. Sampling. Inspection samples of devices required by this specification shall be randomly selected from each inspection lot in accordance with Appendix B-1 of MIL-M-38510.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Microcircuit, Digital, CMOS (Final Logic)	1 OF 1	9292963
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Internal visual (Pre-cap) 2/		100%	3.5.6	MIL-STD-883, Method 2010,B
	High temperature storage (post seal stabilization bake)		100%	3.4,3.5.1 3.5.6	MIL-STD-883, Method 1008,C
	Temperature cycling		100%	3.5.6	MIL-STD-883, Method 1010,C
	Constant acceleration		100%	3.5.6	MIL-STD-883, Method 2001,E Y1 orientation only
	Seal (fine leak)		100%	3.5.6	MIL-STD-883, Method 1014,A or B
	Seal (gross leak)		100%	3.5.6	MIL-STD-883, Method 1014,C
	Electrical parameters @ 25°C (pre burn-in)		100%	3.5.6	Table II, 4.4.3.1 herein
	Burn-in		100%	3.5.6.1	MIL-STD-883, Method 1015,D
	Electrical parameters @ 25°C (post burn-in)		100%	3.5.2	Table II, 4.4.3.1 herein
	External visual		100%	3.3,3.4, 3.5.6,3.6 3.10	MIL-STD-883, Method 2009
	Country of origin		100%	3.11	Visual

NOTES: 1/ Screening. 2/ See 3.8.1: Random monitoring of metallization process by Government inspector.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Microcircuit, Digital, CMOS (Final Logic)				9292963
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Electrical parameters @ -45°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1 herein
	Electrical parameters @ +85°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1 herein
	Oscillator function		LTPD 15%	3.5.4	4.5.14
	Reset pins/bond pads		LPTD 15%	3.5.5	4.5.15

NOTES: 1/ Group A Tests: (see 4.4.3.2, 4.4.3.3).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63321A(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Microcircuit, Digital, CMOS (Final Logic)		9292963
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Subgroup 1				
	Physical dimensions	4	0-1	3.2,3.4	MIL-STD-883, Method 2016
	Subgroup 2				
	Resistance to solvents	3	0-1	3.4	MIL-STD-883, Method 2015
	Bond strength (bond pulls Note 2)		LTPD 15%	3.4	MIL-STD-883, Method 2011,D
	Subgroup 3				
	Solderability (leads inspected Note 3)		LTPD 15%	3.4	MIL-STD-883, Method 2003

NOTES: 1/ Group B Tests (see 4.4.3.4).
2/ The LTPD sample size is the number of bond pulls selected from a minimum number of 10 devices.
3/ The LTPD sample size is the number of leads inspected: In no case shall less than 3 devices be used to provide the number of leads required.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-M-63321A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.4	Microcircuit, Digital, CMOS (Final Logic)	1 OF 1	9292963
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST 1/	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Subgroup 1				
	Steady state life		LTPD 5%	3.5.7	4.5.17
	Electrical parameters @ 25°C ± 5°C		LTPD 5%	3.5.2	Table II, 4.4.3.1
	Subgroup 2				
	Mechanical shock		LTPD 15%	3.4	MIL-STD-883, Method 2002,D
	Vibration, variable frequency		LTPD 15%	3.4	MIL-STD-883, Method 2007,A
	Seal (fine leak)		LTPD 15%	3.4	MIL-STD-883, Method 1014,AorB
	Seal (gross leak)		LTPD 15%	3.4	MIL-STD-883, Method 1014,C
	External visual		LTPD 15%	3.3,3.4 3.5.6, 3.6,3.10	MIL-STD-883, Method 2009
	Electrical parameters @ 25°C ± 5°C		LTPD 15%	3.5.2	Table II, 4.4.3.1
	Packaging		3/	5.1	MIL-M-55565
	Packing		3/	5.1	MIL-M-55565
	Marking		3/	5.2	MIL-M-55565 & MIL-STD-1188

NOTES:
1/ Group C Tests (see 4.4.3.5).
2/ The sample size units shall be subjected to all the tests in a subgroup.
3/ Sampling and inspection as per MIL-M-55565.

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4.4.3 Testing. Testing is described in the first article and quality conformance inspection "Classification of Defects and Tests" forms and below.

4.4.3.1 Electrical parameters @ 25°C ± 5°C testing. Each device shall be tested for compliance with the requirements listed in TABLE II. Any device classified as defective shall be removed from lot.

4.4.3.2 Testing sequence. The order of group testing shall be A, B, and C; that is only lots that passed Group A tests shall be submitted for Group B testing, and only lots that passed Group B tests shall be submitted for Group C testing.

4.4.3.3 Group A testing. Random samples from each inspection lot shall be selected, in accordance with this specification, and tested at the minimum and maximum temperatures for compliance with the requirements listed in Table II. A lot which fail the Group A test shall be classified as defective until all the defective devices within the inspection lot are identified and removed by subjecting all the devices in the lot to the Group A test.

4.4.3.4 Group B testing. Random samples from each inspection lot that passed Group A tests shall be selected and tested for compliance with the requirements in Group B tests in accordance with this specification. An inspection lot shall not be shipped if the devices of any subgroup of the Group B tests fail to comply with all of the specified requirements.

4.4.3.5 Group C testing. Group C tests shall be performed on a continuing schedule of at least once every ninety (90) days while the device is being produced. Random samples from one inspection lot that passed Group B tests shall be selected and tested for compliance with the requirements of the Group C tests in accordance with this specification. Further inspection lots shall not be shipped if the devices of any subgroup of the Group C tests fail to comply with all of the specified requirements.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is identified in the Classification of Defects and Tests Forms and in 4.5 herein. The contractor shall submit for approval inspection equipment designs, including alternate equivalent designs which he may desire to use, in accordance with the terms of the contract. See Section 20 of MIL-M-38510 and 6.3 herein.

4.5 Test methods, circuits and procedures.

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4.5.1 Alternate methods, circuits and procedures. Unless otherwise stated in the specific test methods, the methods, circuits and procedures shown are given as basic testing and measurement methods. For alternate methods, circuits and procedures, the manufacturer shall demonstrate to the procuring activity that the alternates which he may desire and propose to use are equivalent and give results within the desired accuracy of measurement (See 6.3).

4.5.2 Defectives. Any device failing to meet the applicable requirements in the following tests shall be classed and labeled defective and segregated from the inspection lot.

4.5.3 Test conditions. Unless otherwise specified all examinations and tests shall be made under the following conditions.

Temperature	+25°C
Humidity	20 to 60 % RH
Pressure	29 ± 2 inches Hg
Voltage and currents	± 2%
Temperature tolerance	± 5°C

4.5.4 Input current (Pins 2, 3, 4, 5, 6, 7, 9, 10, 11, 13). Connect pins 2, 3, 4, 5, 6, 7, and 13 to V_{DD} and measure the current into each pin. Connect pins 3, 4, 5, 6, 7, 9, 10, 11, and 13 to ground and measure the current out of each pin. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.5 Leakage current (oscillator). Measure the leakage current with the device connected as shown in Figure 10. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.6 Leakage current low. Measure the leakage current in accordance with the test sequence of Table V, and with the device connected as shown in Figure 13. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.7 Input current (Pin 2). Measure the current out of pin 2 when it is connected to the ground. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

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4.5.8 Input current (Pins 9, 10, 11). Measure the current into each pin (9, 10, 11) when it is connected to V_{DD}. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.9 Output low. When the output at pins (1, 12, 14, and 15) are sinking current and held at 1.0 volt, measure the current into each pin. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.10 Output high. When the output at pins (1, 12, 14, and 15) are sourcing current and held at 4.0 volts, measure the current out of each pin. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.11 Switching point. With pin 8 connected to ground and pin 16 connected to 7.5 - 0.1 volts apply 2.5 - 0.10 volts to pin 13, then confirm that the output voltage at pin 12 is less that 1.5 volts. Apply 5.0 + 0.1 volts to pin 13 and confirm that the output voltage at pin 12 is greater than 6.0 volts. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.12 Noise immunity. Noise immunity at input pins (2, 3, 4, 5, 6, 7, 9, 10, 11) shall be confirmed during the functional tests (See 4.5.13). This is a non-destructive test.

4.5.13 Functional tests. The functional tests are to be performed in a manner that confirms the functional timing diagrams defined in Figures 1 thru 5. The tests established for figure 1 shall be performed at the V_{DD} levels of 5.0 and 7.5 volts, and unless otherwise specified, all other functional tests are to be performed at the V_{DD} level of 5.0 volts. Noise immunity requirements shall be confirmed during the functional tests. When performed at the V_{DD} levels of 5.0 and 7.5 volts, the following voltage levels shall be used for logic "0" and logic "1".

Logic "0" = 1.0 volts @ V_{DD} = 5.0 volts

= 1.5 volts @ V_{DD} = 7.5 volts

Logic "1" = 4.0 volts @ V_{DD} = 5.0 volts

= 6.0 volts @ V_{DD} = 7.5 volts

Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

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4.5.14 Oscillator function test. The device shall be connected as shown in Figure 10 and tested for the requirements specified in paragraph 3.5.4. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.15 Dual pins/bond pads. The resistance specified in 3.5.5 shall be measured using a Government approved ohmmeter. During this test, 5.0 $\pm$ .10 volts shall be applied between pin 8 (ground) and pin 16 (V_{DD}), and the drive current from the ohmmeter shall be 100 microamperes maximum. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.16 Burn-in screen. All devices shall be subjected to the burn-in screening in accordance with MIL-STD-883, Method 1015, Condition D (168 hours minimum @ 125°C $\pm$ 5°C) and as specified herein. For this test the device shall be connected as shown in Figure 11 and conditioned at the specified test temperature. The supply voltage V_{DD} applied to pin 16 shall be 7.5 volts and the conditioning signal applied to the inputs shall be as shown in Figure 11. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.17 Steady state life. Devices shall be subjected to the steady state life test in accordance with MIL-STD-883, Method 1005, Condition D, as specified herein. Test duration shall be 1000 hours minimum at +125°C $\pm$ 5°. For this test the device shall be connected as shown in Figure 11; the same test set up as used for the burn-in test. Electrical parameters shall be measured before, at 360 (+160, -0) hours into, and after the steady state life test in accordance with Method 1005 and Table II. Any device failing to meet the applicable requirement shall be classed defective. This is a non-destructive test.

4.5.18 Static discharge (V_{zap}) test. All input device pins having input protection circuitry for static discharge shall be tested in accordance with the procedure specified in figure 12.

4.5.19 Data requirements. The contractor shall submit lot traveler(s) and a quality inspection test, demonstration and evaluation report giving all detailed inspection and test results in accordance with Data Item Description DI-R-1724 (tailored) format to the technical agency listed in 6.5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Devices shall be delivered under this specification shall be prepared for delivery in accordance with the requirements of MIL-M-38510, Level C. Electrostatic discharge (ES) protection in accordance with the requirements of MIL-M-55565 shall be provided.

5.2 Marking. Marking shall be in accordance with MIL-STD-1188.

5.3 Shipping. When microcircuit circuit packages of more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The device covered by this specification is intended for use in the U.S. Army Ordnance.

6.2 Ordering data. The procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-M-38510.
d. Requirement for certification of conformance for each lot or shipment of product.

6.3 Submission of inspection equipment designs for approval. (See 4.4.4). Submit equipment designs as required to:

Commander
ARRADCOM
ATTN: DRDAR-QAA-I
Dover, New Jersey 07801

6.4 Drawings. Drawings listed in section 2 of this specification under the heading U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is not under the cognizance of ARRADCOM.

6.5 Distribution of shop traveler(s), inspection and test results, process control. Shop traveler(s), inspection and test results and process control shall be sent to: Commander, U.S. Army Armament Research & Development Command, ATTN: DRDAR-QAA-M. Dover, New Jersey 07801.

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6.6 Contract data requirements

a. Contract data requirements for inspection equipment designs shall conform to data item description DI-E-1118 or DI-R-1714 with the appropriate addendum to each.

b. Shop traveler(s) shall be submitted.

c. Process control documents shall be submitted in accordance with DI-P-1604 (Mod) with attachment 1 listed on DD Form 1423 incorporated in the contract.

d. Contract data requirements for quality inspection test report in accordance with Data Item Description DI-R-1724 (tailored) giving the detailed test results.

6.7 Changes from previous issue. Due to the extensiveness of the changes, asterisks are not used in this revision to identify changes with respect to the previous issue.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project No: 1345-A-152

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TABLE IV. SIGNAL SUMMARY.

PACKAGE PIN NO.	INPUT/ OUTPUT	SIGNAL SYMBOL
1	Out	F
2	In	$\overline{L}$
3	In	$\overline{C}$
4	In	$\overline{C}$
5	In	Q
6	In	Q
7	In	F_1
8	Supply	V_{SS} (GND)
9	In	B
10	In	A
11	In	Reset
12	Out	F21
13	In	F22
14	Out	F23
15	Out	D
16	Supply	V_{DD}

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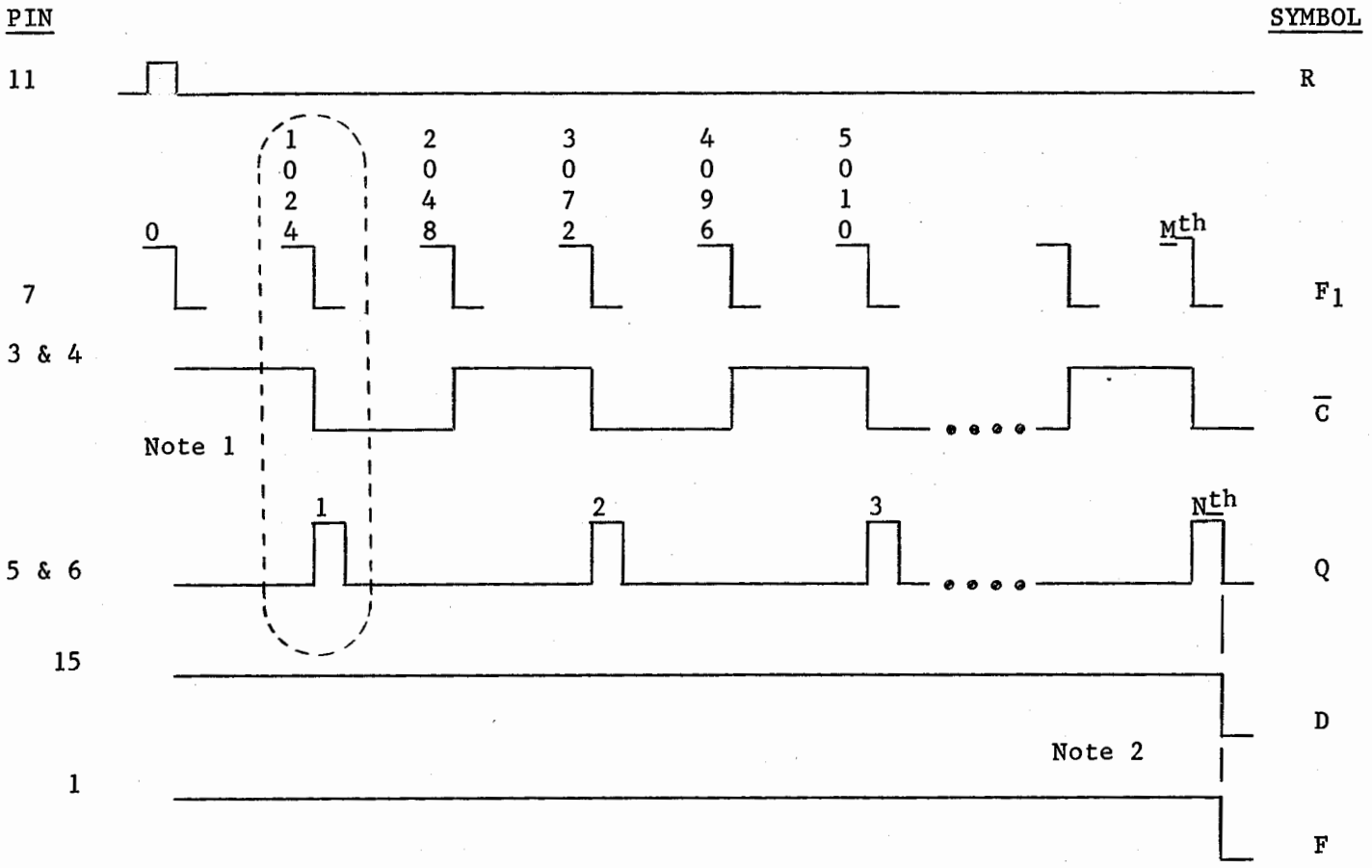


FIGURE 1. Functional timing diagram (self-destruct). The diagram above applies when the self-destruct logic is set as follows:

Logic Levels	N^{th} Q Pulse	M^{th} F_1 Count (Falling Edge)
Pin 10 @ "0" Pin 9 @ "0"	32	64,512
Pin 10 @ "0" Pin 9 @ "1"	64	130,048
Pin 10 @ "1" Pin 9 @ "0"	96	195,584

See Figure 6 for device set up.

Note 1: See Figure 2 for details of the relationship between F_1 and Q input signals.

Note 2: The negative transition of the D and F output signals occurs on the M (th) + 256.5 F_1 count.

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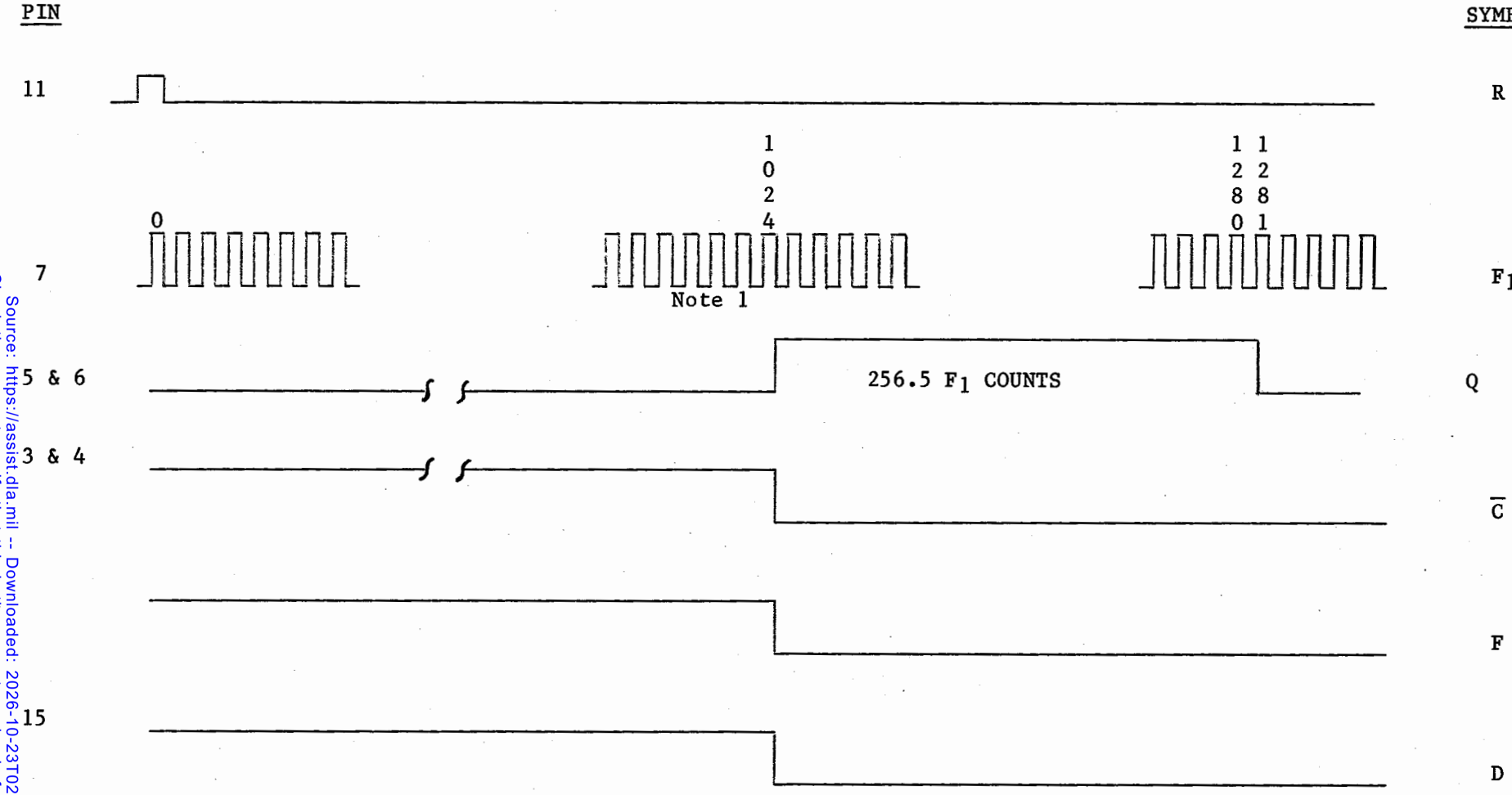


FIGURE 2. Functional timing diagram (self-destruct early). The above applies when the self-destruct logic is set as follows: Pin 10 and Pin 9 set at logic "1".

NOTE 1: Negative transition of F_1 occurs before positive transition of Q.

See figure 6 for test set-up.

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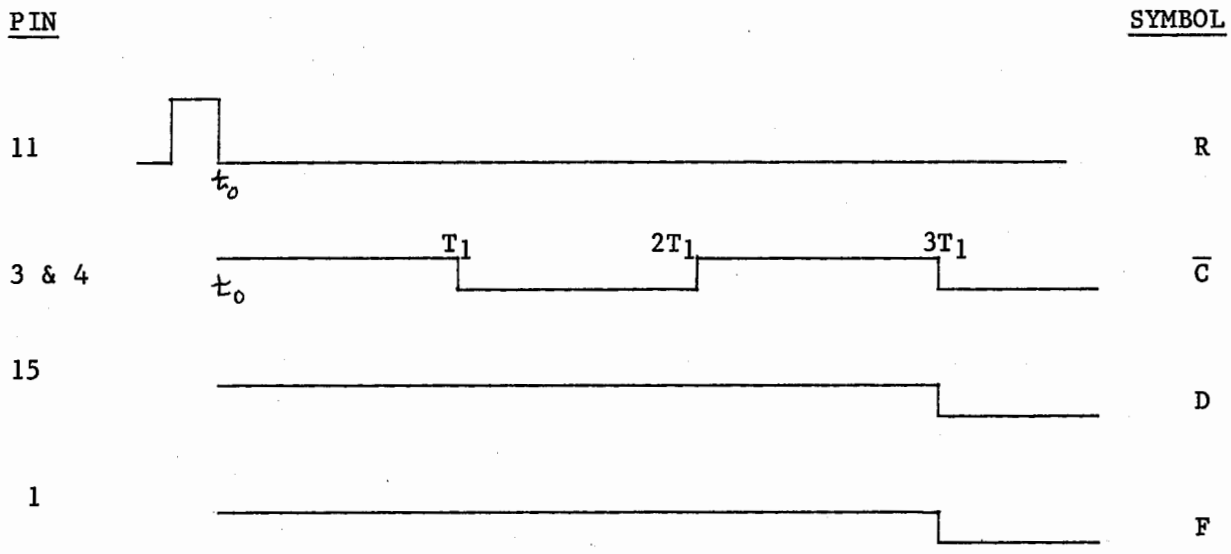


FIGURE 3. Function timing diagram for cycle monitor logic test. T_1 = 150 milliseconds (minimum).

See Figure 7 for device test set-up.

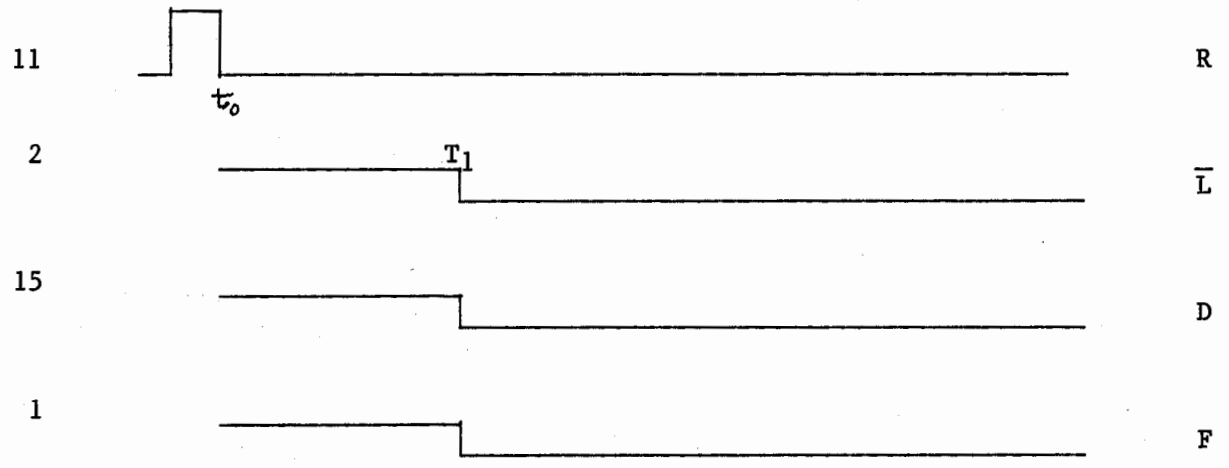


FIGURE 4. Function timing diagram for target signal test. T_1 = 150 milliseconds (minimum).

See Figure 8 for device test set-up.

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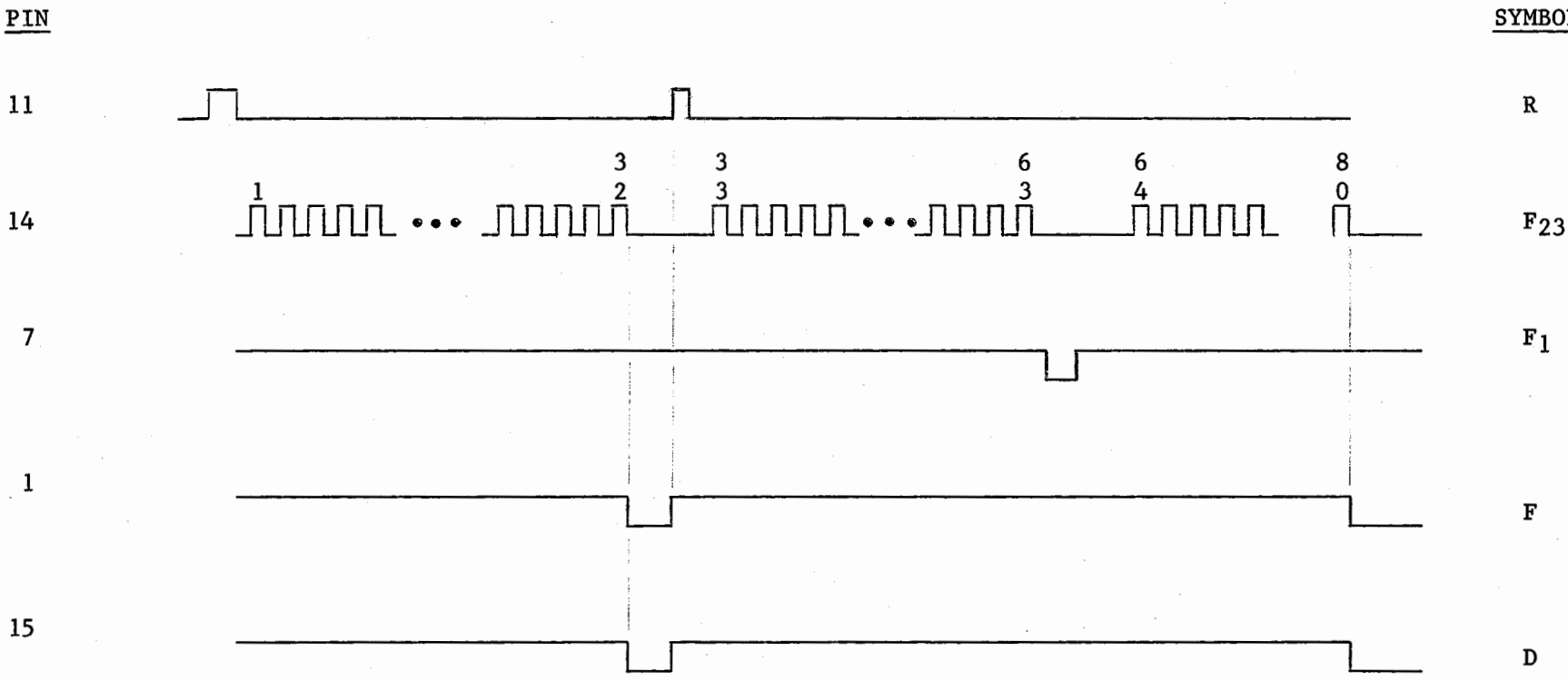


FIGURE 5. Function timing diagram for oscillator check logic test.

See Figure 9 for device test set-up. Input signal F_{22} is applied at Pin 13.

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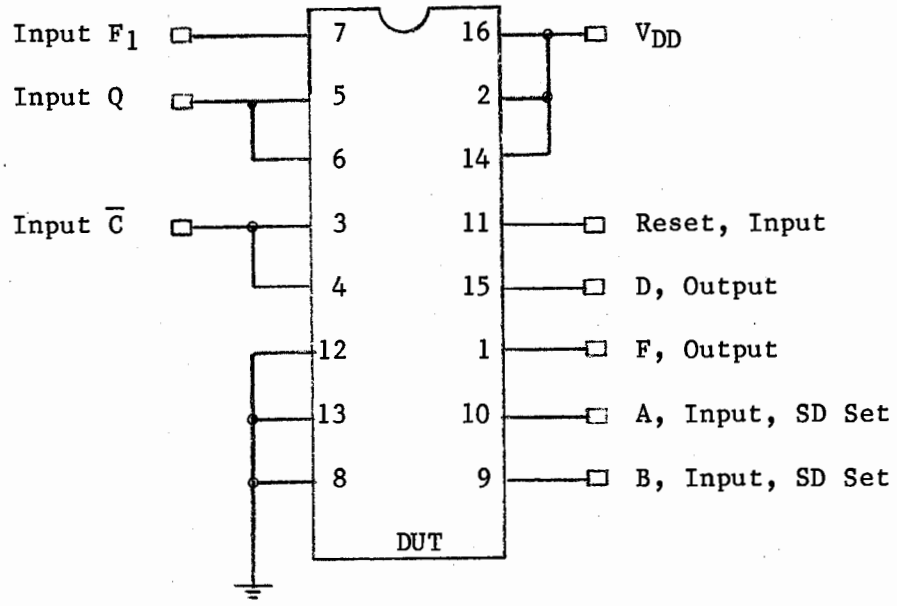


FIGURE 6. Test set up for self-destruct logic test. See Figures 1 and 2 for requirements on the input and output signals.

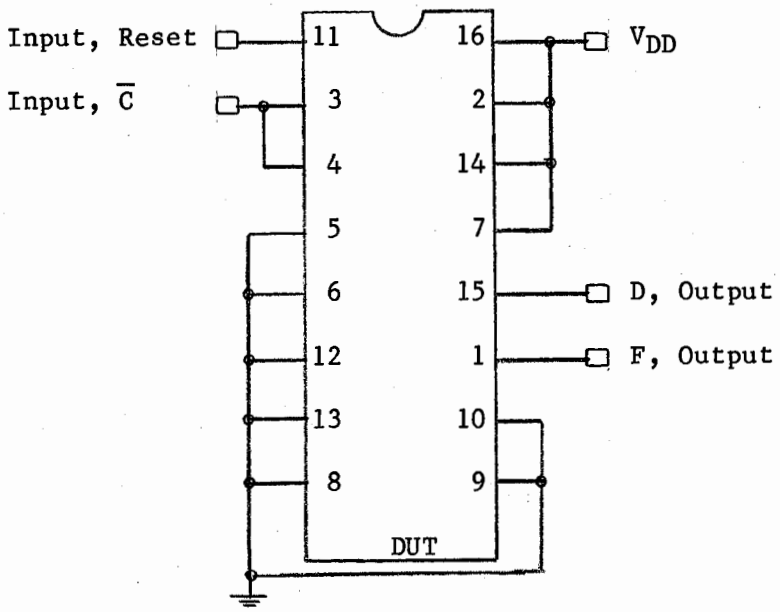


FIGURE 7. Test set up for the cycle monitor logic test. See Figure 3 for the requirements on the input and output signals.

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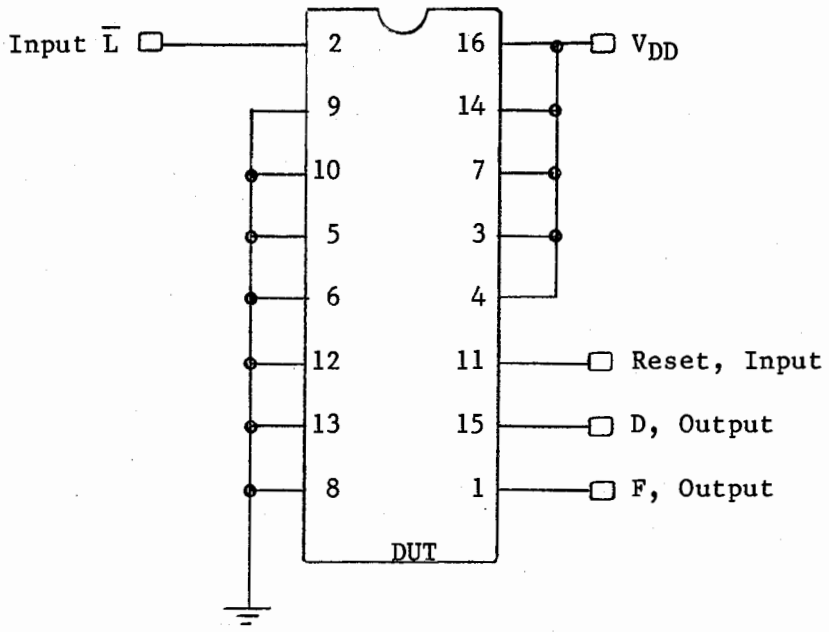


FIGURE 8. Test set up for target signal test. See Figure 4 for requirements on the input and output signals.

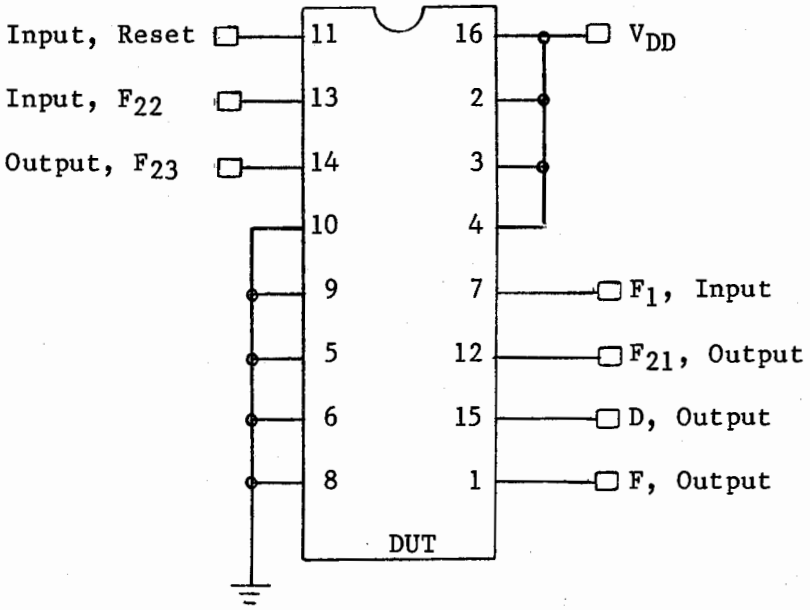
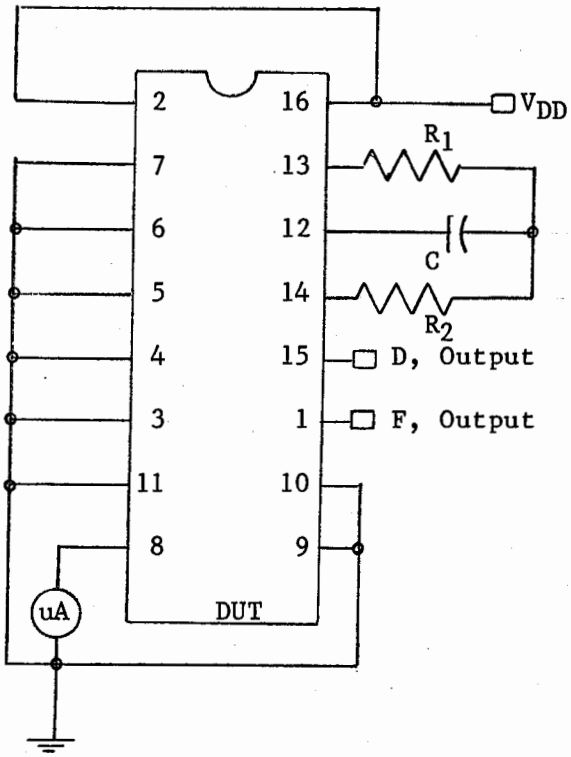


FIGURE 9. Test set up for the oscillator check logic test. See Figure 5 for the requirements on the input and output signals. Output signals F_{23} and F_{21} are the same as input signal F_{22} except that F_{23} is 180° out of phase.

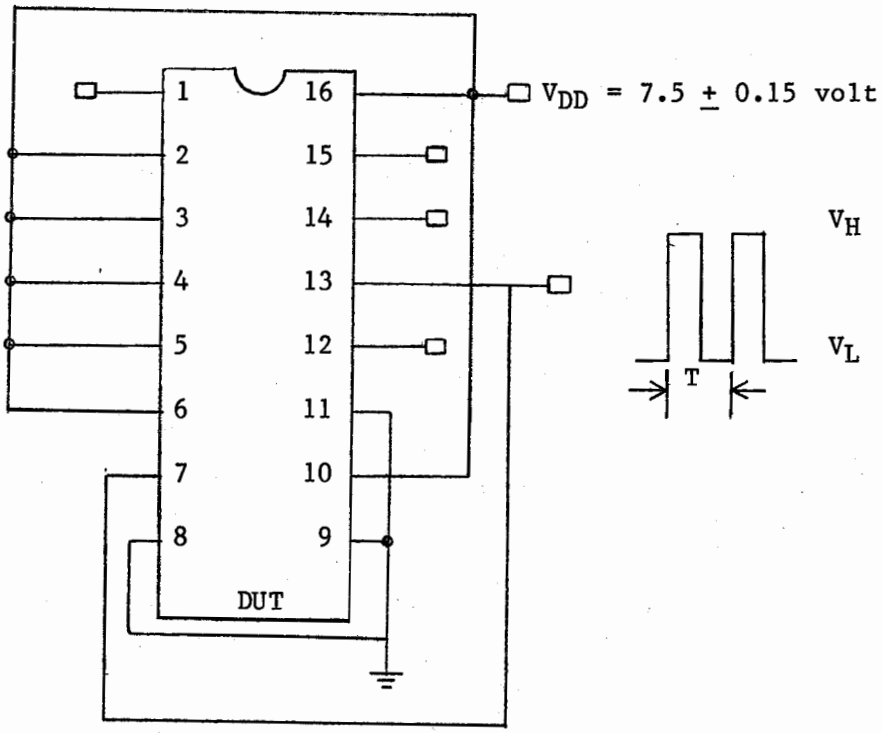
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	Nominal Value	Tolerance	Style	Type	TTC
C	0.01 uF	± 1.0%	Ceramic	NPO	± 25 PPM
R_1	2.0 M-ohms	± 5.0%	Composition		
R_2	470 K-ohms	± 1.0%	Metal Film		± 50 PPM

FIGURE 10. Oscillator test circuit.

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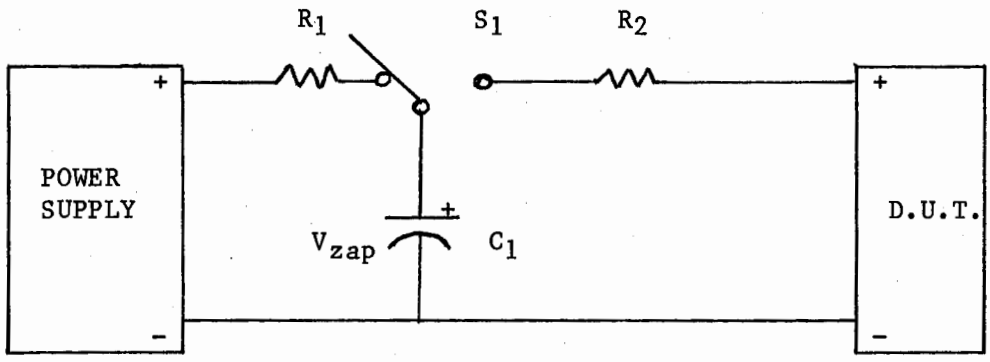
V_H = 7.5 ± -.15 volts

V_L = 0.5 ± 0.15 volts

T = 400 ± 40 uS period, 50% duty factor.

FIGURE 11. Burn-in and steady state of life test circuit.

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R_1 = 10 to 50 M-ohms V_{zap} = 400 V charge on C_1
R_2 = 1.5 K-ohms S_1 = Hg. Wetted "bounceless" relay
C_1 = 100 pF

FIGURE 12. Static discharge test circuit.

Static discharge (V_{zap}) test. Unless otherwise specified, one input terminal of the device under test (DUT) shall be subjected to a voltage pulse from a high voltage source. This destructive test shall be conducted as follows using the test circuit on Figure 12. All input terminals with static discharge protection shall be tested but only one input terminal per device shall be tested.

a.. Measure input current, high level (I_{IH}) and low level (I_{IL}) 25°C ± 5°C at one input terminal of device under test (DUT). Also measure quiescent device current (I_L) at 25°C ± 5°C. The test limit for a single terminal measurement of I_{IH}, and I_{IL}, and I_L shall be increased a maximum of 10% of their value specified in the electrical characteristic Table II.

b. Apply the static discharge (V_{zap}) to the DUT in each of the following modes by charging C1 to V_{zap} with S1 in position 1 and then switching to position 2.

Modes	DUT +Terminal	DUT -Terminal
1	V_{DD}	Input
2	Input	V_{SS}

c. Within 24 hours repeat the I_L, I_{IL} and I_{IH} measurements on the same terminals as performed above. If the DUT exhibits leakage currents in excess of the specified test limit (see a) at this time, it is defective. (Destructive test).

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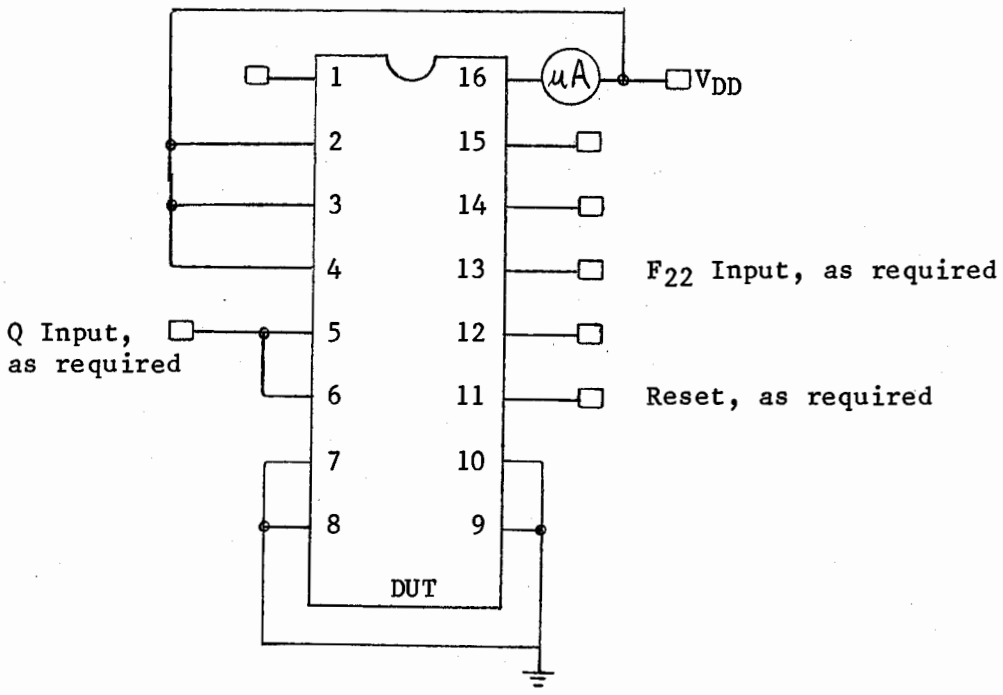


FIGURE 13. Test set up for leakage current low characteristic. Reset and the Q, F_{22} inputs are defined below. The number in ( ) defines the number of pulses (⊓) applied to the specific input during the given sequence.

TABLE V. - Test sequence for leakage current low characteristic.

Sequence No.	Test Condition Logic Levels			Action
	Q	F_{22}	R	
1	0	0	0	
2	0	0	(1)	
3	0	0	0	Measure current
4	(42)	(5)	0	
5	0	0	0	Measure current
6	(43)	(5)	0	
7	0	0	0	Measure current
8	(42)	(5)	0	
9	0	0	0	Measure current

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